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# Ex2: WEAT Bias Audit using Google News Word2Vec

import gensim.downloader as api
import numpy as np
import matplotlib.pyplot as plt
from numpy.linalg import norm

# 1 Load Pre-trained Google News Word2Vec Model
print("Downloading / Loading Google News Word2Vec model...")
model = api.load("word2vec-google-news-300")
print("Model Loaded Successfully!")

# 2 Define Target and Attribute Word Sets

# Gender Target Groups
X = ["man", "male", "boy", "brother", "father", "son"]
Y = ["woman", "female", "girl", "sister", "mother", "daughter"]

# Career vs Family Attributes
A = ["career", "business", "office", "salary", "professional", "corporation"]
B = ["family", "home", "children", "parents", "marriage", "relatives"]

# 3 Remove Words Not Present in Model
X = [w for w in X if w in model]
Y = [w for w in Y if w in model]
A = [w for w in A if w in model]
B = [w for w in B if w in model]

print("Filtered Words Ready")

# 4 Helper Functions

def cosine_similarity(w1, w2):
    return np.dot(w1, w2) / (norm(w1) * norm(w2))

def association(w, A, B):
    mean_A = np.mean([cosine_similarity(model[w], model[a]) for a in A])
    mean_B = np.mean([cosine_similarity(model[w], model[b]) for b in B])
    return mean_A - mean_B

def weat_score(X, Y, A, B):
    return sum([association(x, A, B) for x in X]) - \
        sum([association(y, A, B) for y in Y])

def effect_size(X, Y, A, B):
    assoc_X = [association(x, A, B) for x in X]
    assoc_Y = [association(y, A, B) for y in Y]

    mean_diff = np.mean(assoc_X) - np.mean(assoc_Y)
    pooled_std = np.std(assoc_X + assoc_Y)

    return mean_diff / pooled_std

# 5 Compute WEAT Score and Effect Size
score = weat_score(X, Y, A, B)
effect = effect_size(X, Y, A, B)

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print("\n===== WEAT RESULTS =====")
print("WEAT Score:", round(score, 4))
print("Effect Size:", round(effect, 4))

# 6 Interpretation
if effect > 0:
    print("Interpretation: Target group X is more associated with Attribute A.")
elif effect < 0:
    print("Interpretation: Target group Y is more associated with Attribute A.")
else:
    print("Interpretation: No detectable bias.")

# 7 Visualization
plt.figure()
plt.bar(["Effect Size"], [effect])
plt.axhline(0)
plt.title("WEAT Bias Effect Size (Gender: Career vs Family)")
plt.ylabel("Effect Size")
plt.show()

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Downloading / Loading Google News Word2Vec model...
 [=====] 100.0% 1662.8/1662.8MB downloaded
 Model Loaded Successfully!
 Filtered Words Ready

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===== WEAT RESULTS =====
WEAT Score: 0.2238
Effect Size: 0.2829
Interpretation: Target group X is more associated with Attribute A.

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